

DAA - Assessment Tool 1

Course Code: CSA06 | Course name: Design and Analysis of Algorithms

CO1: Determine algorithm efficiency using asymptotic notations and Master Theorem

Question 1: Online Examination Evaluation System

Recurrence: $T(n) = 3T(n/3) + n^2$ ($a=3, b=3, f(n)=n^2$) $\rightarrow$ Master Theorem Case 3 $\rightarrow T(n) = \Theta(n^2)$

Code:

```
def evaluate_scripts(n, depth=0):
    indent = " " * depth
    if n <= 1:
        print(f"{indent}Base case: n={n}")
        return 1
    print(f"{indent}Splitting n={n} into 3 parts of size {n//3}")
    result = 0
    for i in range(3):
        result += evaluate_scripts(n // 3, depth + 1)
    combine_cost = n * n
    print(f"{indent}Combining at n={n}, cost = n^2 = {combine_cost}")
    return result + combine_cost

print("=== Q1 Execution: T(n) = 3T(n/3) + n^2 ===")
total = evaluate_scripts(9)
print(f"Total simulated cost (Q1): {total}")
```

Output:

```
=== Q1 Execution: T(n) = 3T(n/3) + n^2 ===
Splitting n=9 into 3 parts of size 3
  Splitting n=3 into 3 parts of size 1
    Base case: n=1
    Base case: n=1
    Base case: n=1
  Combining at n=3, cost = n^2 = 9
  Splitting n=3 into 3 parts of size 1
    Base case: n=1
    Base case: n=1
    Base case: n=1
  Combining at n=3, cost = n^2 = 9
  Splitting n=3 into 3 parts of size 1
    Base case: n=1
    Base case: n=1
    Base case: n=1
  Combining at n=3, cost = n^2 = 9
Combining at n=9, cost = n^2 = 81
Total simulated cost (Q1): 117
```

Question 2: AI Model Training Pipeline

Recurrence: $T(n) = 9T(n/3) + n^2$ ($a=9, b=3, f(n)=n^2$) $\rightarrow$ Master Theorem Case 2 $\rightarrow T(n) = \Theta(n^2 \log n)$

Code:

```
def train_model(n, depth=0):
    indent = " " * depth
    if n <= 1:
        print(f"{indent}Base case: n={n}")
        return 1
    print(f"{indent}Splitting n={n} into 9 parts of size {n//3}")
    result = 0
    for i in range(9):
        result += train_model(n // 3, depth + 1)
    combine_cost = n * n
    print(f"{indent}Combining at n={n}, cost = n^2 = {combine_cost}")
    return result + combine_cost

print("=== Q2 Execution: T(n) = 9T(n/3) + n^2 ===")
total = train_model(9)
print(f"Total simulated cost (Q2): {total}")
```

Output:

```
=== Q2 Execution: T(n) = 9T(n/3) + n^2 ===
Splitting n=9 into 9 parts of size 3
Splitting n=3 into 9 parts of size 1
```

[illegible]

```
Combining at n=3, cost =  $n^2 = 9$ 
Splitting n=3 into 9 parts of size 1
  Base case: n=1
  Base case: n=1
  Base case: n=1
  Base case: n=1
  Base case: n=1
  Base case: n=1
  Base case: n=1
  Base case: n=1
  Base case: n=1
Combining at n=3, cost =  $n^2 = 9$ 
Splitting n=3 into 9 parts of size 1
  Base case: n=1
  Base case: n=1
  Base case: n=1
  Base case: n=1
  Base case: n=1
  Base case: n=1
  Base case: n=1
  Base case: n=1
  Base case: n=1
Combining at n=3, cost =  $n^2 = 9$ 
Combining at n=9, cost =  $n^2 = 81$ 
Total simulated cost (Q2): 243
```